IRRIGATION ENGINEERING AND FLOOD CONTROL

Water Requirements of crop

General:

Every crop requires a certain quantity of water after a certain fixed interval, throughout its period of growth. If the natural rain is sufficient and timely so as to satisfy both these requirements, no irrigation water is required for raising that crop. The area where irrigation is a must for agriculture is called the arid region, while the area in which inferior crops can be grown without irrigation is called a semi-arid region.

The term 'Water requirements of a crop' means the total quantity and the way in which a crop requires water, from the time it is sown to the time it is harvested. It is very clear from the above discussion that the water requirement, will vary with the crop as well as with the place. In other words, different crops will have different 'Water requirements, and the same crop may have different water requirement places of the same country ; depending upon the variations in climates, type of soils, methods of cultivation, and useful rainfalls, etc.

Crop period: The time period that elapses from the instant of its sowing to the instant of its harvesting is called the crop-period.

Base Period: The time between the first watering of a crop at the time of its sowing to its last watering before harvesting is called the Base period or the Base of the crop.

Crop period is slightly more than the base period, but for all the practical purposes, they are taken as one and the same thing, and generally expressed in days and will be represented by B (in days).

Duty and Delta of a Crop

Delta: Each crop requires a certain amount of water after a certain fixed interval of time throughout its period of growth. The depth of water required every time, generally varies front to 10 cm depending upon the type of the crop, climate and soil. The time interval between two such consecutive watering is called the frequency of irrigation, or rotation period. The rotation period may vary between 6-15 days for different crops. The summation of the total water depth supplied during the base period of a crop, for its full growth, will evidently represent the total quantity of water required by the crop for its full-fledged nourishment. This total quantity of water required by the crop for its full growth (maturity) may be expressed in hectare-metre (Acre-ft) or in million cubic metres (million cubic-ft) or simply as depth to which water would stand on the irrigated area, if the total quantity supplied were to stand above the surface without percolation or evaporation. This total depth of water (in cm) required by a crop to come to maturity is called its delta (Δ).

Example 2.1. If rice requires about 10 cm depth of water at an average interval of about 10 days, and the crop period for rice is 120 days, find out the delta for rice.

Solution. Water is required at an interval of 10 days for a period of 120 days. It evidently means that 12 no. of waterings are required, and each time, 10 cm depth of water is required.

Therefore, total depth of water required A =12 x 10 cm= 120 cm.

Hence A for rice = 120 cm. Ans.

Example 2.2. If wheat requires about 7.5 cm of water after every 28 days, and the base period for wheat is 140 days, find out the value of delta for wheat.

Solution. Assuming the base period to be representing the crop period, as per usual practice, we can easily infer that the water is required at an average interval of 28 days up to a total period of 140 days.

This means that 140÷28 = 5 no. of waterings are required.

The depth of water required each time = 7.5 cm.

Total depth of water reqd. in 140 days = 5 x 7.5 cm = 37.5 cm

Hence, Δ for wheat = 37.5 cm. Ans.

Duty of Water:

The 'duty' of water is the relationship between the volume of water and the area of the crop it matures. It may be defined as the number of hectares of land irrigated for full growth of a given crop by supply of 1 m^3/sec of water continuously during the entire base period (B) of that crop. The duty is generally represented by the letter D.

Relation between duty and delta:

Let there be a crop of base period B days. Let one cumec of water be applied to this crop on the field for B days. Now, the volume of water applied to this crop during B days

V= (1 x 60 x 60 x 24 x B) m^3 = 86 400 B (m^3)

By definition of duty (D), one cubic meter supplied for B days matures D hectares of land. This quantity of water (V) matures D hectares of land. Or 10^4 D sq. m of area.

Total depth of water applied on this land

$$= \text{Volume} \div \text{Area} = \frac{86400B}{10^4 D} = \frac{8.64B}{D} \text{ metres}$$

By definition, this total depth of water is called delta (Δ).

$$\Delta = \frac{8.64B}{D} \text{ metres}$$

$$\Delta = \frac{864B}{D} \text{ cm}$$

Where; Δ is in cm, B is in days, and D is duty in hectares/cumec.

Example 2.3. Find the delta fort crop when its duty is 864 hectares/cumec on the field, the base period. of this crop is 120 days

Solution:

In this question, B = 120 days and D = 864 hectares/curnec

$$\Delta = \frac{864B}{D} \text{ cm} = 120 \text{ cm: Ans.}$$

Duty at various places.

In a large canal irrigation system, the water from its source first of all flows into the main canal ; from the main canal, it flows into the branch canal ; from the branch canal, it flows into the distributary ; from the distributary, it flows into the minor ; and then into the field channels (water-courses) ; and finally into the fields. A systematic layout of a canal system is shown in Fig. 2.1. During the passage of water from these irrigation channels, water is lost due to evaporation and percolation. These losses are called Transit losses or Transmission or. Conveyance losses in channels.

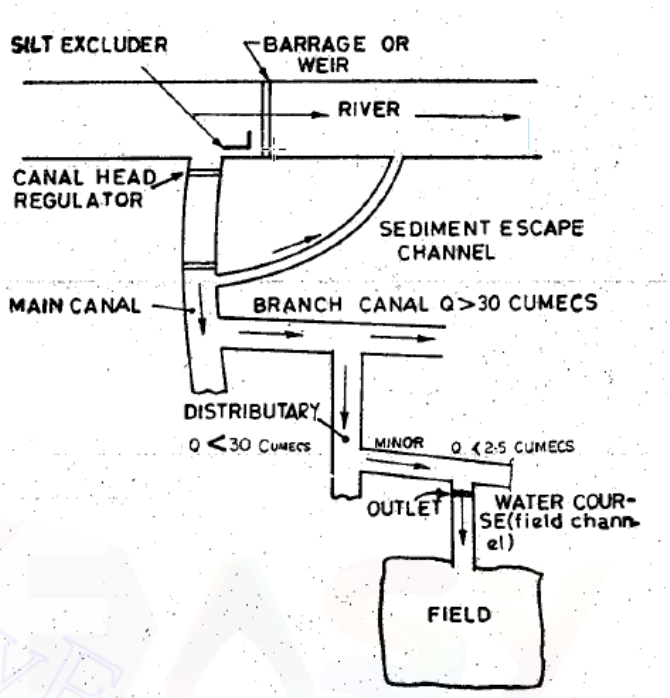


Fig. 2.1. Layout of a canal system.

Factors on which duty depends:

Duty of irrigation water depends upon the following factors:

(ii) Type of crop

(ii) Climate and season.

(iii) Useful rainfall

(iv) Type of soil.

(v) Efficiency of cultivation method.

Importance of duty:

It helps us in designing an efficient canal irrigation system. Knowing the total available water at the head of a main canal, and the overall duty for all the crops required to be irrigated in different seasons of the year, the area which can be irrigated can be worked out. Inversely, if we know the crops area required to be irrigated and their duties, we can work out the discharge required for designing the channel.

Crop Seasons:

From the agricultural point of view, the year can be divided into two principal cropping seasons, i.e. Rabi and Kharif": Normally, Rabi starts from 1st October and ends on. 31st March ; while Kharif starts from 1st April and ends on 30th September. These dates are not rigid dead lines. The time may vary up to 1-3 months on either side: Sugarcane, which is an important cash crop; extends over both seasons. The Kharif crops are rice, bajra, jowar, maize, cotton, tobacco, groundnut, etc. The Rabi crops are wheat, barley, gram, linseed, mustard, potatoes, etc. Kharif crops are also called 'summer crops' and Rabi crops as 'Winter crops'. Kharif crops require about two to three times the quantity of water required by the Rabi crops.

Certain Important Definitions:

Kharif-Rabi ratio or Crop ratio: The area to be irrigated for Rabi crop is generally more than that for the Kharif crop. This ratio of proposed areas, to be irrigated in Kharif season to that in the Rabi season is called, Kharif-Rabi ratio. This ratio is generally 1 : 2

Paleo irrigation: Sometimes, in the initial stages before the crop is sown, the land is very dry. This particularly happens at the time of sowing of Rabi crops because of hot September, when the soil may be too dry to be sown easily. In such a case, the soil is moistened with water, so as to help in sowing of the crops. This is known as Paleo irrigation.

Kor-watering: The first watering which is given to a crop, when the crop is a few centimetres high, is called kor-watering. The optimum depth of kor-watering for different crops are different. For example, the optimum depth of kor-watering for Rice is 19 cm. for Wheat is about 13.5 cm, and for Sugarcane is 16.5 cm. The kor-period for rice varies from 2 to 4 weeks, and that for wheat varies from 3 to 8 weeks.

Cash crops: A cash crop may be defined as a crop which has to be encashed in the market for processing, etc. as it cannot be consumed directly by the cultivators. All non-food crops are thus, included in cash crops. Crops like jute, tea, cotton, tobacco, sugarcane, etc. are, therefore, called cash crops. The food crops like wheat, rice, barley, maize, etc. are excluded from the list of cash crops

Crops rotation: When the same crop is grown again and again in the same field, the fertility of land gets reduced as the soil becomes deficient in plant foods favorable to that particular crop. In order to enhance the fertility of the land and to make the soil regain its original structure, it is often found necessary and helpful to give some rest to the land. This can be achieved either by allowing the land to lie fallow without any cultivation for some time, or to grow crops which do not mainly require those salts or foods which were mainly required by the earlier grown crop. This method of growing different crops in rotation, one after the other, in the same field, is called Rotation of Crops.

Optimum Utilization of Irrigation Water:

If a crop is sown and produced under absolutely identical conditions, using different amounts of water depths, the yield is found to vary. The yield increases with water, reaches a certain maximum value and then falls down, as shown in Fig. 2.2. The quantity of water at which the yield is maximum, is Called the optimum water depth.

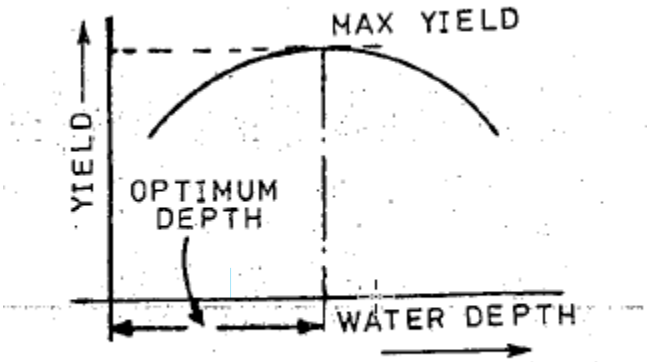


Fig. 2.2.

Irrigation Efficiencies:

Efficiency is the ratio of the water output to the water input, and is usually expressed as percentage, there are different kinds of irrigation efficiencies, as given below :

(i) **Efficiency of water conveyance**: It is the ratio of the water delivered into the fields from the outlet point of the channel, to the water entering into the channel at its starting point. It may be represented by η_c.

(ii) **Efficiency of water-application**: It is the ratio of the quantity of water stored into the root zone, of the crops to the quantity of water actually delivered into the field. It may be represented by η_a. It may also be called on farm efficiency, as it takes into consideration the water lost in the farm.

(iii) **Efficiency of water-storage**: It is the ratio of the water stored in the root zone during irrigation to the water needed in the root zone prior to irrigation (i.e. field capacity – existing moisture content). It may be represented by η_s

(iv) **Efficiency of water use**: It is the ratio of the water beneficially used, including leaching water, to the quantity of water delivered. It may be represented by η_u.

Example 2.4. One cumec of water is pumped into a farm distribution system. 0.8 cumec is delivered to a turn-out, 0.9 kilometer from the well. Compute the conveyance efficiency.

Solution. By definition $\eta_c = \frac{0.8}{1} \times 100\% = 80\%$

Example 2.5. 10 cumecs of water is delivered to a 32 hectare field, for 4 hours. Soil probing after the irrigation indicates that 0.3 metre of water has been stored in the root zone. Compute the water application efficiency.

Solution. Volume of water supplied by 10 cumecs of water applied for 4 hours = (10 x 4 x 60 x 60) m^3 = 1,44,000 m^3 = 14.4 x 104m^3 = 14.4m x 104 m^2=14.4 ha.m

Input =14.4 ha.m

Output= 32 hectares land is storing water upto 0.3 m depth.

Output = 32 x 0.3 ha.m = 9.6 ha.m

Water application efficiency η_a = $\frac{9.6}{14.4} \times 100 = 66.67\%$

(v) **Uniformity coefficient or Water distribution efficiency**. The effectiveness of irrigation may also be measured by its water distribution efficiency (η_d), which .is defined below

$$\eta_d = \left(1 - \frac{d}{D}\right)$$

where η_d = Water distribution efficiency.

D= Mean depth of water stored during irrigation.

d= Average of the absolute values of deviations from the mean.

Example 2.6. The depths of penetrations along the length of a border strip at 30 meter apart were probed: Their observed values are 2.0, 1.9, 1.8, 1.6 and 1.5 metres. Compute the water distribution efficiency.

Solution. The observed des at five stations are 2.0, 1.9, 1.8, 1.6 and 1.5 metres, respectively.

Mean depth = D = 1.76 metres

Values of deviations from the mean are (2.0- 1.76), (1.9 - 1.76), (1.8 - 1.76), (1.6 - 1.76), (1.5 -1.76) i.e. 0.24, 0.14, 0.04, 0.16 and 0.26.

The absolute values of these deviations from the mean |D - observed value| , are. 0.24, 0.14, 0.04, 0.16 and 0.26. The average of these absolute values of deviations from the mean d = 0.84

The water distribution efficiency $\eta_d = \left(1 - \frac{d}{D}\right)$ = 0.905, Ans.

Example 2.7. A stream of 130 litres per second was diverted from a canal and 100 litres per-second were delivered to the field. An area of 1.6 hectares was irrigated in 8 hours. The effective depth of root zone was 1.7 m. The runoff loss in the field was 420 cu.m. The depth of water penetration varied linearly from 1.7 m at the head end of the field to 1.1 m at the tail end. Available moisture holding capacity of the soil is 20 cm per metre depth of soil. It is required to determine the water conveyance efficiency, water application efficiency, water storage efficiency, and water distribution efficiency. Irrigation was started at a moisture extraction level of 50% of the available moisture.

Solution.

(i) Water conveyance efficiency (η_c) = (Water delivered to the fields /Water supplied into the canal at the head) x 100 % = 77% Arts.

$\frac{100}{130} \times 100\%$

(ii) Water application efficiency (η_a) = Water stored in the root zone during irrigation / Water delivered to the field x 100%

Water supplied to field during 8 hours @ 100 litres per second = 100.x 8 x 60 x 60 litres= 2880-cu.m

Runoff loss in the field = 420 cu. m. the water stored in the root zone = 2880 – 420 = 2460 cu. m

Water application efficiency (η_a) = 85.4 % Ans.

(iii) Water storage efficiency (η_s) = Water stored in the root zone during irrigation / Water needed in root zone prior to irrigation x 100%

Moisture holding capacity of soil = 20 cm per m depth x 1.7 m depth of root zone = 34 cm.

Moisture already available in root zone at the time of start of irrigation =0.50 x 34 = 17 ctn.

Additional water required in root zone = 34 – 17 = 17 cm.

The volume of required water = 0.17 x1.6 x 10^4 = 2720 cu.m

But actual water stored in root zone = 2460 cu. m

Water storage efficiency (η_s) = 90 % (say) Ans.

(iv) The water distribution efficiency $\eta_d = \left(1 - \frac{d}{D}\right)$

D= mean depth of water, stored in the root zone. =1.14 m

d= Average of the absolute values of deviations = 0.3

$\eta_d = \left(1 - \frac{d}{D}\right)$ = 0.786 or 78.6% -Ans.

Consumptive use or Evapotranspiration (C_u):

Consumptive use for a particular Crop may be defined as the total amount of water used by the plant in transpiration (building of plant tissues, etc.) and evaporation from adjacent soils or from plant leaves, in any specified time.

Effective Rainfall (Re): Precipitation falling during the growing period of a crop that is available to meet the evapotranspiration needs of the crop, is called effective rainfall

Consumptive Irrigation Requirement (CIR): It is the amount of Irrigation water required in order to meet the evapotranspiration needs of the crop during its full growth.

C.I.R. = Cu – Re

Net Irrigation Requirement (NIR): It is the amount of irrigation water required in order to meet the evapotranspiration need of the crop as well as other needs such as leaching.

Therefore, N.I.R. = Cu - Re + Water lost as percolation in satisfying other needs such as leaching.

Factors Affecting Consumptive Use: Consumptive use or evapotranspiration depends upon all those factors on which evaporation and transpiration depend; such as temperature, sunlight, humidity, wind movement, etc,

Estimation of Consumptive Use Although various methods have been developed in order to estimate evapotranspiration (consumptive use) values of different crop in an area, or for areas vegetated with the same cropping-pattern but the most simple and-commonly-used methods are-:

(1) Blaney-Criddle Equation, and

(2) Hargreave`s class A pan evaporation method.

(3) Penman's equation.

Soil-Moisture-Irrigation Relationship:

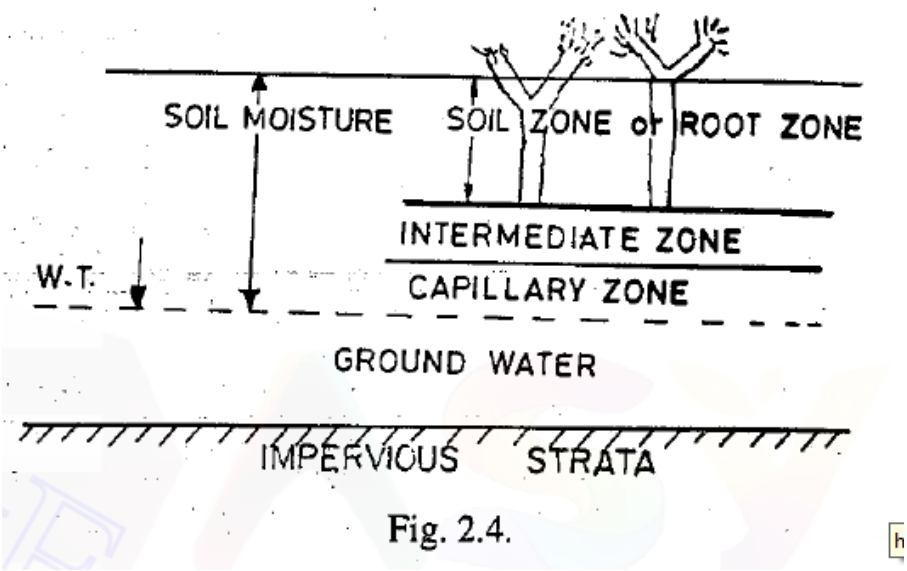


Fig. 2.4.

Field Capacity: Field capacity is the amount of water retained in the soil after excess gravitational water has drained away, usually within 2–5 days after rainfall or irrigation.

Immediately after a rain or irrigation water application, when all the gravity water has drain down to the water table, a certain amount of water is retained on the surfaces of soil grains by molecular attraction and by loose chemical bonds (i.e. adsorption). This water cannot be easily drained under the action of gravity, and is called the field capacity. The field capacity is thus the water Content of soil after free drainage has taken place for a sufficient period. This period of free gravity drainage is generally taken as 2 to 5 days.

The field capacity = (Wt. of water retained in a certain vol. of soil / Wt. of the same volume of dry soil) x 100

If we consider 1 m^2 area of soil and d metre depth of root zone, then the volume of soil is d x 1 = d cubic metres. If the dry unit wt. of soil is Υ_d kN/m^3, then the Wt. of d cubic metres of soil is Υ_d .d kN.

If F is the field capacity, then

$$F = \frac{\text{Wt.of water retained in unit area of soil}}{\Upsilon_d .d}$$

Wt. of water retained in unit area of soil = Υ_d .d.F kN/m^2

Vol. of water stored in unit area of soil = $\frac{\Upsilon_d \text{ .d.F kN/m2}}{\Upsilon_w \text{ kN/}m^3}$kN/m3

or Total water storage capacity of soil in (m depth of water)

$d_w = \frac{\Upsilon_d \text{ .d.F}}{\Upsilon_w}$ m

Yw = 9.81 kn/m^3
Yw = 1000 kgf/m^3
Yw= 62.4 lb/ft^3

Where F = the field capacity m.c.

d = depth of root zone in m

Υ_d = the dry unit wt. of soil.

Υ_w = the unit wt. water

Hence, the depth of water stored in the root zone in filling the soil up to field capacity = $\frac{\Upsilon_d \text{ .d.F}}{\Upsilon_w}$ m

The plants can extract water from the soil till the permanent wilting point is reached. The permanent wilting point is that water content at which plant can no longer extract sufficient water for its growth, and wilts up. It is the point at which permanent wilting of plant take place. It therefore becomes evident that the water which is available to the plants is the difference of field capacity water and permanent wilting point water. This is known as available moisture or maximum storage capacity of soil. Hence, the available water or available moisture may be defined as the difference in water content of the soil between field capacity and permanent wilting point. The water left in the soil after the permanent wilting point is reached, cannot be removed, and is known as, unavailable moisture or Hygroscopic water (See Fig. 2.5).

Readily available moisture: It is that portion of the available moisture which is most easily extracted by the plants, and is approximately 75 to 80% of the available moisture.

Soil-moisture deficiency: The water required to bring the soil moisture content of a given soil to its field capacity is called the field moisture deficiency or soil-moisture deficiency.

It is the amount of water required to bring the soil moisture content up to its field capacity.

Equivalent moisture: Just as the field capacity is the water retained by a saturated soil after being acted upon by gravity; similarly, equivalent moisture is the water retained by a saturated soil after being centrifuged for 30 minutes by a centrifugal force of 1000 times that of gravity. Therefore, it is slightly less, or at the most equal to the field capacity.

Equivalent Moisture: It is the amount of water retained by saturated soil after centrifuging for 30 minutes at 1000 times the force of gravity.

Estimating Depth and Frequency of Irrigation on the Basis of Soil Moisture Regime Concept:

Water or soil moisture is consumed by plants through their roots. It, therefore, becomes necessary that sufficient moisture remains available in the soil from the surface to the root zone depth. As explained earlier, the soil moisture in the root zone can vary between field capacity (upper limit) and wilting point moisture content (lower limit) as shown in Fig. 2.5.

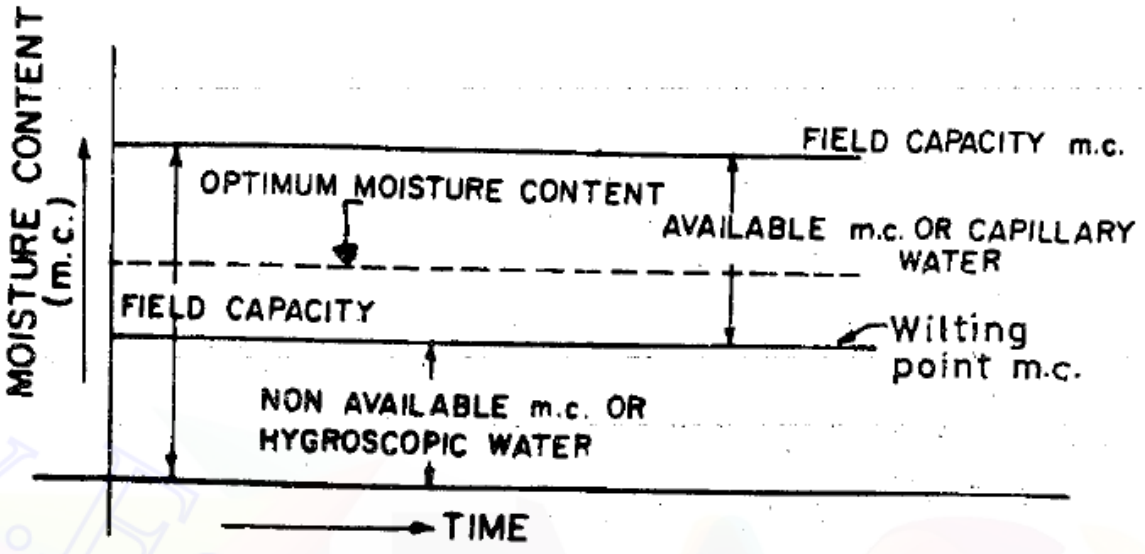


Fig. 2.5.

It is also evident from the previous discussion that the soil moisture is not allowed to be depleted up to the wilting point, as it would result in considerable fall in crop yield. The optimum level, up to which the soil moisture may be allowed to be depleted in the root zone without fall in crop yield, has to be worked out for every crop and soil, by experimentation. The irrigation water should be supplied as soon as the moisture falls up to this optimum level (fixing irrigation frequency) and its quantity should he just sufficient to bring the moisture content up to its field-capacity, making allowance for application losses (thus fixing water depth). Water will be utilized by the plants after the fresh irrigation dose is given, and soil moisture will start falling. It will again be recouped by a fresh dose of irrigation, as soon as the soil moisture reaches the optimum level, as shown in Fig. 2.6.

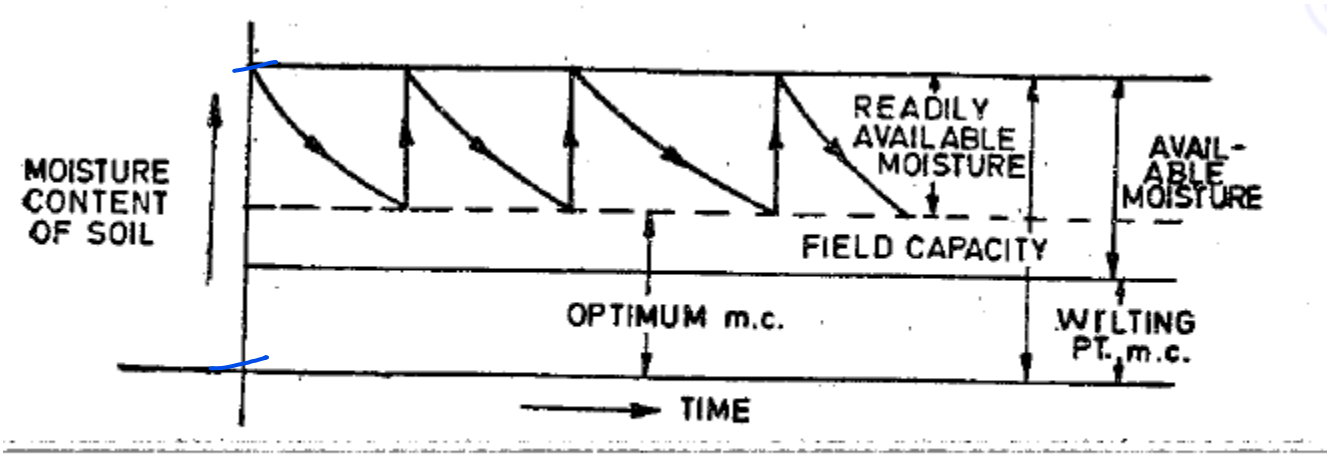


Fig. 2.6.

Example 2.15. After how many days will you supply water to soil in order to ensure sufficient irrigation of the given crop, if

(i) Field capacity of the soil = 28%
(ii) Permanent wilting point = 13%
(iii) Dry density of soil =.1.3 gm/c.c.
(iv) Effective depth of root zone = 70 cm
(v) Daily consumptive use of water for the given crop = 12 mm. Assume any other data, not given.

Solution. We know, by definition of available moisture, that the available moisture = Field capacity— Permanent wilting = 28 - 13 = 15%.

Let us assume that the readily available moisture or the optimum soil moisture level is 80% of available moisture. i.e; readily available moisture = 0.80 x 15% = 12%

মাটির Moisture Content 28% থেকে
কমতে কমতে 16% হলে আমরা irrigation দেব। তখন 16% থেকে আবার 28% পর্যন্ত পানি পূরণ করব।

Optimum moisture =28 - 12=16%

It means that the moisture will be filled by irrigation between 16% and 28%.

Depth of water stored in root zone between these two limits

$d_w = \frac{\Upsilon_d}{\Upsilon_w}$.d. [Field capacity m.c. — Optimum m.c.]

(1.3*.70/1)*.12

Where, $\frac{\Upsilon_d}{\Upsilon_w}$ = 1.3, d= 0.7 m

d_w = 0.1092m= 10.92 cm.

Hence, water available for evapotranspiration = 10.92 cm.

1.2 cm of water is utilized by the plant in 1 day

So, 10.92 cm of water will be utilized by the plant in = 9.1 days; Say 9 days

Hence, after 9 days, water should be supplied to the given crop. Ans.

Example 2.16. Wheat is to be grown in a field having a field capacity equal to 27% and the permanent wilting point is 13%. Find the storage capacity in 80 cm depth of the soil, if the dry unit weight of the soil is 14.72 kN/m^3. If irrigation water is to be supplied when the average soil moisture falls to 18%, find the water depth required to be supplied to the field if the field application efficiency is 80%. What is the amount of water needed at the canal outlet if the water lost in the water-courses and the field channels is 15% of the outlet discharge?

Solution.

Maximum storage capacity or Available moisture = $\frac{\Upsilon_d}{\Upsilon_w}$.d. [Field capacity m.c. — Witting pt. m.c.]

where γ_d = 14.72 kN/m^3 , γ_w = 9.81 kN/m^3

d= depth of root zone = 0.8 m

Max. Storage capacity or max. Available moisture = 0.168 metres =16.8 cm. Ans.

Since the moisture is allowed to vary between 27% and 18%, the deficiency created in this fall

$= \frac{14.72}{9.81}$ x 0.8 [0.27 — 0.18) = 1.2 x 0.09 = 0.108 metres = 10.8 cm.

Hence, 10.8 cm depth of water is the net irrigation requirement.

Quantity of water required to be supplied to the field (F.I.R.) = $\frac{NIR}{\eta_a}$ = 13.5 cm. Ans.

Quantity of water needed at the canal outlet= $\frac{FIR}{\eta_c}$ = 15.88 cm. Ans. ($\eta_c = 0.85$)

Example 2.17. 800 m^3 of water is applied to a farmer's rice field of 0.6 hectares. When the moisture content in the soil falls to 40% of the available water between the field capacity (36%) of soil and permanent wilting point (15%) of the soil crop combination, determine the field application efficiency. The root zone depth of rice is 60 cm. Assume porosity = 0.4

Solution. We have defined Field Capacity m.c. (F) as:

$$F = \frac{\text{Wt.of water contained in a certain vol.of soil}}{\text{Wt.of the same volume of dry soil (t e.wt.of dry soil retaining that water)}}$$

$F = \frac{\gamma_d . V_v}{\gamma_w . V}$ But $\frac{V_v}{V} = n$

So, $\frac{\gamma_d}{\gamma_w} = 1.11$

Max. Quantity of water stored between field capacity (FC) and permanent wilting point (P.W)

$= \frac{\gamma_d}{\gamma_w} d (W_F - W_p) = 0.14$ m

Deficiency of water created when irrigation is done = 60% x 0.14 m = 0.084 m

[Irrigation water is applied when m.c. falls to 40% of m.c. available between F.C. and P.W.]

Hence, irrigation water is supplied to fill up 0.084 m depth of water.

Vol. of irrigation water required to fill up the created deficiency = 0.084 m x (0.6 hect.) = 0.084 m x (0.6 x 10,000) m^2 = 504 m^3.

(i) Actual irrigation water supplied = 800 m^3

(ii) Efficiency of field application = 504/800 = 63% Ans.

Example 2.19. A sandy loam soil holds water at 140 mm/m depth between field capacity and permanent witting point. The root depth of the crop is 30 cm and the allowable depletion of water is 35%. The daily water use by the crop is 5 mm/day. The area to be irrigated is 60 ha and water can be diverted at 281.p.s. The surface irrigation application efficiency is 40%. There are no rainfall and ground water contribution. Determine (i) allowable depletion depth between irrigations. (ii) Frequency of irrigation (iii) net application depth of water (iv) volume of water required (v) time to irrigate 4 ha plot

Total available moisture = 4.2 cm
এর 35% ব্যবহার হলে = 1.47 cm
এই 1.47 cm পানি শেষ হওয়ার আগেই আবার irrigation দিতে হবে।

Solution. Moisture holding capacity of soil = 140 mm/m depth

Depth of root zone = 30 cm = 0.3 m

Moisture holding 'capacity of root zone = 140 x 0.3 m = 42 mm = 4.2. cm

Allowable depletion = 35%

(i)Available moisture depth or Allowable depletion depth between irrigations = 35% x 4.2 cm = 1.47 cm. Ans.
Daily use of water = consumptive use = 5 mm/day

(ii)Frequency of irrigation = Available moisture /Moisture consumed per day = 1.47/0.5 = 2.94 days., say 3 days. Ans.

(ii)Net water depth to be applied while irrigating each time after 3 days = 3 x 0.5 = 1.5 cm Ans. (in place of 1.47 cm)

Field Irrigation requirement = Net irrigation requirement/ efficiency of irrigation = 1.5/0.4 = 3.75 cm

(iv) Qty. of water regd. in the fields = 3.75 cm of water depth = 3.75 cm x Area of field = 3.75 cm x 60 ha

= 0.0375 m x (60 x 10^4) m^2 = 22,500 m^3

Hence, vol. of water reqd. to irrigate 60 ha area, each time at 3 days interval = 22,500 m3 Ans.

(vi) Time to irrigate 4 ha when irrigation water is supplied @ 28 1ps :

Vol. of water reqd. to irrigate 4 ha plot = 3.75 cm x 4 ha = 0.0375x (4 x 10^4) m^3= 1500 m^3

Time during which 1500 m3 of water can be supplied @ 28 1ps. = 14.88 hr. Ans.

Example 2.20. Determine the field capacity of a soil for the following data:

(i) Depth roar zone = 1.8 m
(ii) Existing moisture = 8%
(iii) Dry density of soil = 1450 kg/m^3
(iv) Quantity of water applied to soil = 650 m^3,
(v) Water lost due to deep percolation and evaporation = 10%
(vi) Area to be irrigated = 1000 m^2

Solution. Volume of total water applied = 650 m^3.

Water wasted = 10% of 650 m^3 = 65 m^3

Water used in raising m.c up to field capacity = 650 - 65 = 585 m^3.

Depth of Water used in raising m.c up to field capacity from the existing 8% = 585/1000 = 0.585 m

But water depth required in root zone of depth increase m.c, is given by eqn.

$d_w = \frac{\gamma_d}{\gamma_w}$.d. [Fc — lower limit mc, as fractions.]

Fc = 0.144 = 14.4%

Hence, Field capacity =14.4% Ans.